LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: The Cell

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style questions, and Level 3 is Foundation-level challenge.
2. For every organelle, learn three things: what it looks like, where it is, and what it does.

LEVEL 1: CONCEPTS

Q1. Who discovered cells, and how?

Q2. Which organelle is called the 'powerhouse of the cell', and why?

Q3. Why are lysosomes called 'suicide bags'?

Q4. Name the organelle where proteins are made.

Q5. Which two organelles are found in plant cells but not in animal cells?

LEVEL 2: BOARD-STYLE QUESTIONS

Q6. Give three differences between prokaryotic and eukaryotic cells.

Q7. What is osmosis? What happens to a cell placed in (a) a hypotonic solution and (b) a hypertonic solution?

Q8. What is plasmolysis? When does it happen?

Q9. Give the functions of: (a) the plasma membrane, (b) the endoplasmic reticulum, (c) the Golgi apparatus.

Q10. Why are plastids important to plants? Name the two types.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. What would happen to a cell if its plasma membrane ruptured or broke down?

Q12. What would happen to a cell if it had no Golgi apparatus?

Q13. Raisins kept in water swell up, and fresh grapes kept in strong sugar solution shrink. Explain both using osmosis.

~ Draw a plant cell and an animal cell side by side and label at least eight parts! ~